Deriving Expressions and Formulae - Mark scheme

Question 1 *Practice question written for this worksheet (not from a past paper)*

Ali is x years old. Beth is 5 years older than Ali.

Cara is twice as old as Ali. The sum of their ages is 61.

Work out Ali's age.

$x + (x + 5) + 2x = 61$

$4x = 56$

Answer: 14

Question 2 *Practice question written for this worksheet (not from a past paper)*

12 tiles and one bag of adhesive costing 30 dollars cost 90 dollars altogether.

Each tile costs x dollars. Work out x.

$12x + 30 = 90$

$12x = 60$

Answer: 5

Question 3 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 12*

Q	Working	Answer	Mark	Notes
	$C - 5$ oe **or** $2C$ oe **or** T = a linear expression in C		3	M1 for one of $C - 5$ oe **or** $2C$ oe **or** T = linear expression in C
	$C + C - 5 + 2C$ $(= 4C - 5)$ oe **or** for T = an expression in C with the expression in C coming from adding at least 2 of C, $2C$, $C - 5$ eg $T = 2C + C - 5$ or $T = C + C^2 + C - 5$			M1
		$T = 4C - 5$		A1 oe but must be simplified eg allow $T = 4 \times C - 5$
				Total 3 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2022 · Question 12

Q	Working	Answer	Mark	Notes
	eg $3x - 24 + 102 - x = 180$ oe **or** $90 + 90 + 3x - 24 + 102 - x = 360$ oe		3	M1 for setting up a correct equation
	eg $2x = 180 - 78$ oe **or** $2x = 360 - 258$ oe **or** eg $(180 + 24 - 102) \div 2$ **or** $360 - (90 + 90 - 24 + 102) \div 2$			M1 for simplifying and isolating the x term **or** for a complete calculation to find the value of x
		51		A1
				Total 3 marks

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 13

Q	Working	Answer	Mark	Notes
(a)	$2n$ **or** $n + 7$		2	M1 for $2n$ oe **or** $n + 7$ oe
	Correct answer scores full marks (unless from obvious incorrect working)	$4n + 7$		A1 oe eg $7 + 4n$
(b)		$T - "(4n + 7)"$	1	B1ft oe, ft T – [their answer (a)]
				Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 14

		$F = 2r + 7h$	3	B3 for $F = 2r + 7h$ oe (B2 for $2r + 7h$ **or** $F = 2r + ah$ **or** $F = br + 7h$ **or** $F = 2h + 7r$) (B1 for $2r + ah$ **or** $br + 7h$ **or** $2h + 7r$ **or** $F = 2r$ **or** $F = 7h$ **or** for $F =$ an incorrect expression in r and h)
	Correct answer scores full marks (unless from obvious incorrect working)			**Total 3 marks**

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 6

6		$T = 6m + 9g$	3	B3 Or $T = 3(2m + 3g)$ [award B2 if $T = 6m + 9g$ is incorrectly simplified](condone $T = 6 \times m + 9 \times g$) if not B3 then B2 for $T = 6m + kg$ **or** $T = km + 9g$ (k may be zero) **or** $6m + 9g$ if not B2 then B1 for $6m$ **or** $9g$ **or** $T = am + bg$ (where $a \neq 0$ or 6 and $b \neq 0$ or 9)
				Total 3 marks

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Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 13

13	$\frac{60}{360}\times 2\times\pi\times r$ oe or $\frac{1}{6}\times 2\times\pi\times r$ oe		3	M1	for finding the length of the arc
	"$\frac{60}{360}\times 2\times\pi\times r$"$+2r$ oe or "$\frac{1}{6}\times 2\times\pi\times r$"$+2r$ oe			M1	dep on M1 for a complete expression from correct working for a method for the perimeter
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$P=r\left(\frac{1}{3}\pi+2\right)$		A1	oe eg $P=r(0.33...\pi+2)$ or $P=\left(\frac{1}{3}\pi+2\right)r$ or $P=\left(2+\frac{120}{360}\pi\right)r$ or $P=\left(\frac{120}{360}\pi+2\right)r$
					Total 3 marks

Question 9

Practice question written for this worksheet (not from a past paper)

A rectangle has length 3 cm more than its width w cm. Its area is 54 cm².

(a) Show that $w^2 + 3w - 54 = 0$

(b) Work out the width of the rectangle.

(a) $w(w+3) = 54$, so $w^2 + 3w - 54 = 0$

(b) $(w+9)(w-6) = 0$; w must be positive

Answer: w = 6 cm

Question 10

Practice question written for this worksheet (not from a past paper)

A rectangle has length $(2x + 1)$ cm and width $(x + 4)$ cm.

Write expressions, in simplest form, for (a) its area (b) its perimeter.

(a) $(2x+1)(x+4) = 2x^2 + 8x + x + 4$

(b) $2(2x + 1 + x + 4) = 2(3x + 5)$

Answer: (a) $2x^2 + 9x + 4$ (b) $6x + 10$

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 2

Q	Working	Answer	Mark	Notes
	$1-(0.20+0.26)\ (=0.54)$ oe or $x+2x+0.26+0.20=1$ oe or $x+2x=0.54$ oe or		4	M1 showing clear understanding that the total of probabilities is 1 If probabilities are given as percentages then % sign must be seen
	$\frac{"0.54"}{3}(=0.18)$ or $\frac{2}{3}\times"0.54"(=0.36)$ oe or $0.54\times450\ (=243)$			M1 for a correct method to find x or $2x$
	$(2\times)$ "0.18" × 450 oe or 81 or "0.36" × 450 oe			M1 or for $\frac{81}{450}$ or $\frac{162}{450}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	162		A1
				Total 4 marks

ALT	$(0.2\times450)+(0.26\times450)\ (=207)$ oe or $90+117\ (=207)$ or $0.46\times450\ (=207)$		4	M1
	$450-$ "207" $(=243)$			M1
	$\frac{1}{3}\times"243"$ or 81 or $\frac{2}{3}\times"243"$			M1 or for $\frac{81}{450}$ or $\frac{162}{450}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	162		A1
				Total 4 marks

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Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2025 · Question 15

Q	Working	Answer	Mark	Notes
(a)	$n + 7$ **or** $3n$ **or** $4n + 7$		2	M1 for sight of $n + 7$ **or** $3n$ **or** $4n + 7$ for $3n$, allow $3 \times n$ or $n3$ but do **not** allow this mark for $3n$ if n^3 also seen (mark as choice) Allow expressions to be seen within a sum, difference, product or quotient
	Correct answer scores full marks (unless from obvious incorrect working)	$n + n + 7 + 3n$		A1 oe eg $2n + 7 + 3n$ **or** $5n + 7$ ISW if correct expression seen but then simplified incorrectly Condone a formula, eg $T = n + n + 7 + 3n$ but do **not** allow $n = n + n + 7 + 3n$ for A1
(b)	eg (Sid =) $72 \div 3$ (= 24) or (Erin =) $\frac{k}{3} + 7$		2	M1 for correct method to find the number of counters Sid has **or** a correct expression for how many counters Erin has in terms of how many counters Kabir has (allow use of any letter in place of k, including n)
	Correct answer scores full marks (unless from obvious incorrect working)	31		A1
				Total 4 marks

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CHALLENGE QUESTIONS

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 12

Q	Working	Answer	Mark	Notes
	$9x + 27 + 2x - 23 = 180$ oe eg $11x + 4 = 180$ **or** $\frac{180-27+23}{9+2}(=16)$ **or** $\frac{180-4}{11}(=16)$		4	M1 for using the angle sum on a line to form an equation in x **or** for a numerical method to find x using the angle sum on the line
	$2\times"16"-23\ (=9)$ **or** $9\times"16"+27\ (=171)$			M1 for using the value of x to find the exterior or interior angle
	eg $\frac{360}{n}="9"$ oe **or** $(n=)\frac{360}{"9"}$ $\frac{180(n-2)}{n}="171"$ oe **or** $(n=)\frac{360}{180-"171"}$ oe			M1 for using the formula for an exterior or an interior angle to form an equation in n **or** for a complete method to find n
	Correct answer scores full marks (unless from obvious incorrect working)	40		A1
				Total 4 marks

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Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 23

Q	Working	Answer	Mark	Notes
	$(7x-9)-(x+2)=(x+2)-(4x-14)$ oe eg $6x-11=16-3x$ **OR** $x+2=4x-14+d$ and $7x-9=4x-14+2d$ oe eg $-3x+16=d$ and $3x+5=2d$		4	M1 for setting up an equation in x **OR** two simultaneous equations in x and d
$x=3$ **and** $a=-2$ **and** $d=7$ **OR** $x=3$ **and** eg $(S_{40}=)\frac{40}{2}\left[2(4x-14)+39(-3x+16)\right]$ **or** $x=3$ **and** eg $(S_{40}=)\frac{40}{2}\left[2(4x-14)+39(6x-11)\right]$				M1 correct values or values from correct substitution **OR** $x=3$ **and** S_{40} expressed in terms of x allow (40 – 1) for 39
$(S_{40}=)\frac{40}{2}(2\times"-2"+39\times"7")$ **or** eg $(S_{40}=)\frac{40}{2}\left[2(4\times"3"-14)+39(-3\times"3"+16)\right]$ **or** $(S_{40}=)\frac{40}{2}\left[2(4\times"3"-14)+39(6\times"3"-11)\right]$				M1 allow use of their a **and** their d **or** their x as long as clearly stated allow (40 – 1) for 39
	Working required	5380		A1 (dep on M1)
				Total 4 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 23

Question	Working	Answer	Mark	Notes
	$\frac{12}{x}\times\frac{x-12}{x-1}$ oe **or** $\frac{x-12}{x}\times\frac{12}{x-1}$ oe		5	M1 for a correct expression for P(P,Y) or P(Y,P)
	$2\left(\frac{12}{x}\times\frac{x-12}{x-1}\right)=\frac{18}{47}$ oe **or** $\frac{12}{x}\times\frac{x-12}{x-1}+\frac{x-12}{x}\times\frac{12}{x-1}=\frac{18}{47}$ oe			M1 for a correct equation for $P(P,Y)+P(Y,P)=\frac{18}{47}$ This M mark implies first M mark
	$18x^2-1146x+13536(=0)$ oe **or** $9x^2-573x+6768(=0)$ oe **or** $3x^2-191x+2256(=0)$ oe			M1 dep on M2 simplified to a 3 term quadratic (ax^2+bx+c (=0) in any form) with at least 2 coefficients (a or b or c) are correct
	$(3x-47)(x-48)(=0)$			M1 ft dep on first M1 for solving *their* 3-term quadratic

	or $x=\frac{--191\pm\sqrt{(-191)^2-(4\times3\times2256)}}{2\times3}$ or $3\left[\left(x-\frac{191}{6}\right)^2-\left(\frac{191}{6}\right)^2\right]+2256(=0)$ $\left(x+\frac{-191}{2\times3}\right)^2-\left(\frac{-191}{2\times3}\right)^2+\frac{2256}{3}(=0)$				equation using any method – if factorising, allow brackets which expand to give 2 out of the 3 terms correct. If using the formula or completing the square allow one sign error and some simplification as far as $(x=)\frac{191\pm\sqrt{36\,481-27\,072}}{6}$ or $3\left(x-\frac{191}{6}\right)^2-\frac{9409}{12}(=0)$ Award this mark for answers of 48 and $\frac{47}{3}$ dep on M3
	Working required	48	48		A1	cao, dep on first 3 M marks. Do not award if non-integer solution also given.
						Total 5 marks

	$\frac{12}{x}\times\frac{x-12}{x}$ oe or $\frac{x-12}{x}\times\frac{12}{x}$ oe			5	M0
	$2\left(\frac{12}{x}\times\frac{x-12}{x}\right)=\frac{18}{47}$ oe or $\frac{12}{x}\times\frac{x-12}{x}+\frac{x-12}{x}\times\frac{12}{x}=\frac{18}{47}$ oe				M1 for a correct equation for P(P,Y) + P(Y,P) = $\frac{18}{47}$

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						This M mark DOES NOT imply the first M mark
	$18x^2 - 1128x + 13536 (= 0)$ oe or $9x^2 - 564x + 6768 (= 0)$ oe or $3x^2 - 188x + 2256 (= 0)$ oe				M1	dep on previous M1 simplified to a 3 term quadratic ($ax^2 + bx + c$ (=0) in any form) with at least 2 coefficients (a or b or c) are correct
	$(x =) \dfrac{--188 \pm \sqrt{(-188)^2 - (4 \times 3 \times 2256)}}{2 \times 3}$ or $3\left[\left(x - \dfrac{188}{6}\right)^2 - \left(\dfrac{188}{6}\right)^2\right] + 2256 (= 0)$ oe or $(x =) \dfrac{188 \pm 4\sqrt{517}}{6}$ or $\left(x + \dfrac{-188}{2 \times 3}\right)^2 - \left(\dfrac{-188}{2 \times 3}\right)^2 + \dfrac{2256}{3} (= 0)$ or $(x =)$ 46(.491756) and 16(.174910)				M1	ft dep on second M1 for solving *their* 3-term quadratic equation using any method – if factorising, allow brackets which expand to give 2 out of the 3 terms correct. If using the formula or completing the square allow one sign error and some simplification as far as $(x =) \dfrac{188 \pm \sqrt{35344 - 27072}}{6}$ or $3\left(x - \dfrac{188}{6}\right)^2 - \dfrac{2068}{3} (= 0)$ Award this mark for both values of x dep on M2
	Working required				A0	
						Total 3 marks

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Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 23

23	eg $(y+3)(x+2)=4x(3y-5)$ **or** $\frac{3y-5}{x+2}=\frac{y+3}{4x}$ **or** $\frac{x+2}{3y-5}=\frac{4x}{y+3}$ **or** $\frac{3y-5}{y+3}=\frac{x+2}{4x}$ **or** $\frac{4x}{x+2}=\frac{y+3}{3y-5}$		5	M1	for correct use of intersecting chords theorem to form an equation
	eg $xy+2y+3x+6=12xy-20x$ oe			M1	for expanding the brackets **or** for removing the fractions **and** expanding the brackets, allow one error in one term we can **ft** $4x(y+3)=(x+2)(3y-5)$ oe for the 2nd, 3rd and 4th method marks
	eg $20x+3x+6=12xy-xy-2y$ oe **or** $23x+6=11xy-2y$ oe			M1ft	dep on previous M1 for correctly collecting all the y terms on one side and non-y terms on the other side
	eg $23x+6=y(11x-2)$ **or** $20x+3x+6=y(12x-x-2)$			M1ft	dep on 2nd M mark for factorising, for y, an equation in the form $ax+b=cxy+dy$ (may not be simplified) the factorisation must be correct
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{23x+6}{11x-2}$		A1	oe
					Total 5 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 24

Q	Working	Answer	Mark	Notes
	eg $(d=)(3y-4)-(2x+5)(=3y-2x-9)$ **or** $(2x+5)+d=3y-4$ oe **or** $(d=)(4x-2)-(3y-4)(=4x-3y+2)$ **or** $(3y-4)+d=(4x-2)$ **or** $(2d=)(4x-2)-(2x+5)(=2x-7)$ **or** $(2x+5)+2d=(4x-2)$ oe		6	M1 for a correct expression or equation using the common difference, may be in terms of d for this mark we will allow an expression for $-d$ or $-2d$
	eg $216=\frac{9}{2}\left[2(2x+5)+(9-1)d\right]$ **or** $216=\frac{9}{2}\left[2(2x+5)+(9-1)"(3y-2x-9)"\right]$ **or** $216=\frac{9}{2}\left[2(2x+5)+(9-1)"(4x-3y+2)"\right]$ **or** $216=\frac{9}{2}\left[2(2x+5)+(9-1)"(x-3.5)"\right]$ **or** $216=\frac{9}{2}\left[2(2("d+3.5")+5)+8d\right]$			M1 for a correct equation for the sum of 9 terms in x and d **or** in terms of x and y **or** in terms of x or in terms of d for "$3y-2x-9$ we will allow $(3y-4)-(2x+5)$ or for using their incorrect simplification from $(3y-4)-(2x+5)$ shown for "$4x-3y+2$we will allow $(4x-2)-(3y-4)$ or for using their incorrect simplification from $(4x-2)-(3y-4)$ shown similarly for their "$x-3.5$" and their "$d+3.5$"

	eg $6x-6y=-11$ oe **and** $12y-6x=55$ oe **or** $6x-6y=-11$ oe **and** $18x-12y=11$ oe **or** $2x-2d=7$ oe **and** $8d+4x=38$ oe	eg $d=x-3.5$ oe **and** $48=12x-18$ oe **or** $x=d+3.5$ oe **and** $48=12d+24$ oe **or** $d=4.75-0.5x$ oe **and** $3x-2=14.5$ oe			
	one equation must be from the differences and one equation must be from the sum of the terms			M2	**left hand column** 2 correct equations in terms of x and y in the form $px+qy=r$ oe **or** 2 correct equations in terms of x and d in the form $px+qd=r$ oe **right hand column** 2 correct equations in terms of x and y where one is substituted into the other to get a correct equation in the form $px+q=r$ or $py+q=r$ **or** 2 correct equations in terms of x and d where one is substituted into the other to get a correct equation in the form $px+q=r$ or $pd+q=r$ **If not M2 then M1 for one correct equation in any of the required forms from the left hand or right hand column**
	Working required	$x=\frac{11}{2}$ $y=\frac{22}{3}$		A2	(dep on M2) oe (allow 7.3(33…)) (A1 for $x=\frac{11}{2}$ **or** $y=\frac{22}{3}$)
					Total 6 marks

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